SOS: DEEP-SAR OIL SPILL DATASET

Abstract

Deep-SAR Oil Spill (SOS) dataset was produced and shared by HPSCIL of China University of Geosciences, Wuhan and RS-IDEA of Wuhan University. It is constructed to fully advance the oil spill detection task. SOS dataset provides a benchmark resource for the development of state-of-the-art algorithms for SAR image oil spill detection or other related tasks.

1. Deep-SAR Oil Spill (SOS) dataset

The SOS dataset has two study areas, the Gulf of Mexico oil spill area and the Persian Gulf oil spill area. Fig. 1 shows the location of the oil spill in the Gulf of Mexico and a typical subset of the selected ALOS satellite images. The location of the oil spill in the Persian Gulf and a typical subset of the selected Sentinel-1A satellite images are given in Fig. 2. For this reason, data enhancement techniques were used to extend the original dataset. Cropping, rotating, adding noise and other operations are taken on the original grayscale maps. Finally, a total of 3101 images from the Mexico oil spill area were used for training and 776 images were used for testing. A total of 3354 images from the Persian Gulf oil spill area were used for training and 839 images were used for testing.

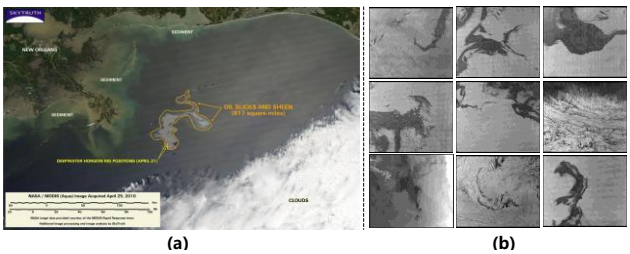


Fig.1. The location of the oil spill in the Gulf of Mexico. (a) NASA/MODIS satellite image taken April 25 showing oil slicks from Deepwater Horizon disaster. (Source: John Amos 2010 [1]) (b) Typical subset of the selected ALOS satellite images.

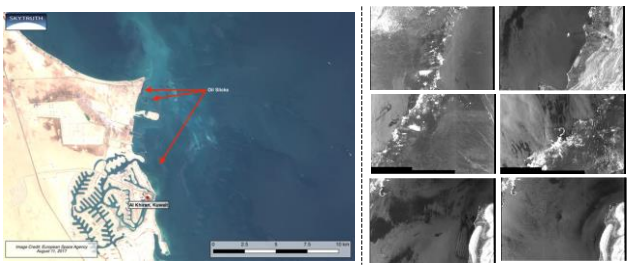


Fig.2. The location of the oil spill in the Persian Gulf. (a) Sentinel-2A multispectral satellite image showing oil slick making landfall along Kuwait's coast near Al Khiran on August 11, 2017. (Source: John Amos 2017 [2]). (b) Typical subset of the selected Sentinel-1A satellite images.

[1] "Gulf Oil Spill Covers 817 Square Miles," SkyTruth, Apr. 26, 2010. [Online]. Available: https://skytruth.org/2010/04/gulf-oil-spill-covers-817-square-miles/ (accessed Mar. 22, 2021).
[2] "Satellite Imagery Reveals Scope of Last Week's Oil Spill in Kuwait," SkyTruth, Aug. 15, 2017. [Online]. Available: https://skytruth.org/2017/08/satellite-imagery-reveals-scope-of-last-weeks-massive-oil-spill-in-kuwait/ (accessed Mar. 19, 2021).

2. Download

SOS dataset tree is as follows:

```
SOS
└———train
|     └———gt
|     └———images
└———val
|     └———gt
|     └———images
```

3. Copyright

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4. Contact

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